



Technische
Universität
Braunschweig

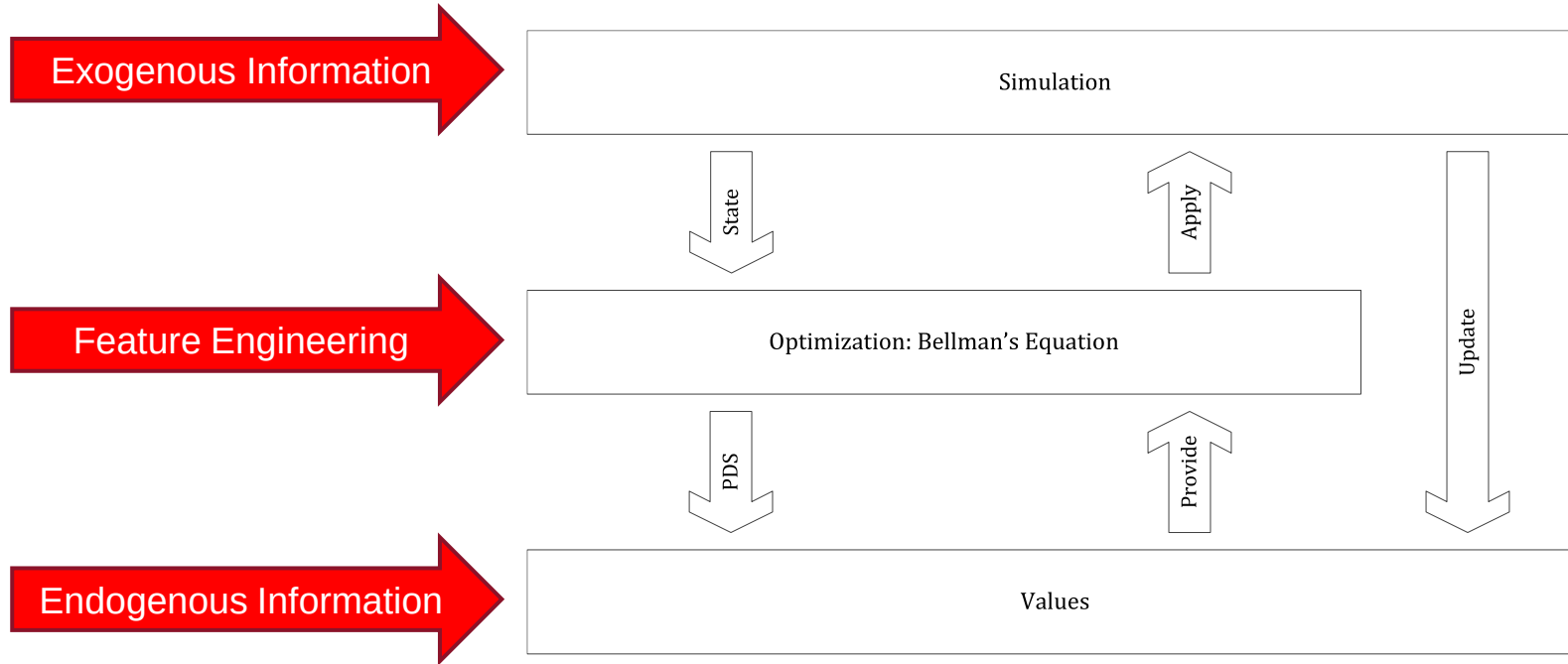
**Decision
Support**
Institut für Wirtschaftsinformatik



Data Driven Decision Making

Lecture 9 – Analytics Issues in Approximate Dynamic Programming

Data Analytics Issues for ADP Methods



Unsupervised vs. Supervised Learning

- Machine Learning differentiates concepts in (un-)supervised learning approaches
- **Unsupervised learning** looks at data objects consisting of attributes only
- It then builds a model that describes all data objects in the best possible way
- It stands proxy for descriptive modeling, an illustrative concept is clustering
- **Supervised learning** looks at data objects consisting of attributes and (one) variable
- It aims at a mapping of attribute values to one distinct value of the variable
- It learns a suitable mapping from training data, later on it discriminates test data
- It stands proxy for predictive modeling, an illustrative concept is classification
- Attribute/variable types determine e.g. regression (numerical) vs. classification (categorical)

Data Objects

We often assume that the data set is provided in form of a simple table

unsupervised learning

	Attribute 1	...	Attribute m
Instance 1	$value_{11}$	$value_{1...}$	$value_{1m}$
...	$value_{...1}$	$value_{.....}$	$value_{...m}$
Instance n	$value_{n1}$	$value_{n...}$	$value_{nm}$

Berthold et al. (2010), Ch. 4.1

supervised learning

	Attribute 1	...	Variable
Instance 1	$value_{11}$	$value_{1...}$	$value_1$
...	$value_{...1}$	$value_{.....}$	$value_{...}$
Instance n	$value_{n1}$	$value_{n...}$	$value_n$

phase
training
training
test data

The rows of the table are called **instances, records, samples or data objects**

The columns of the table are called **attributes, features or variables**

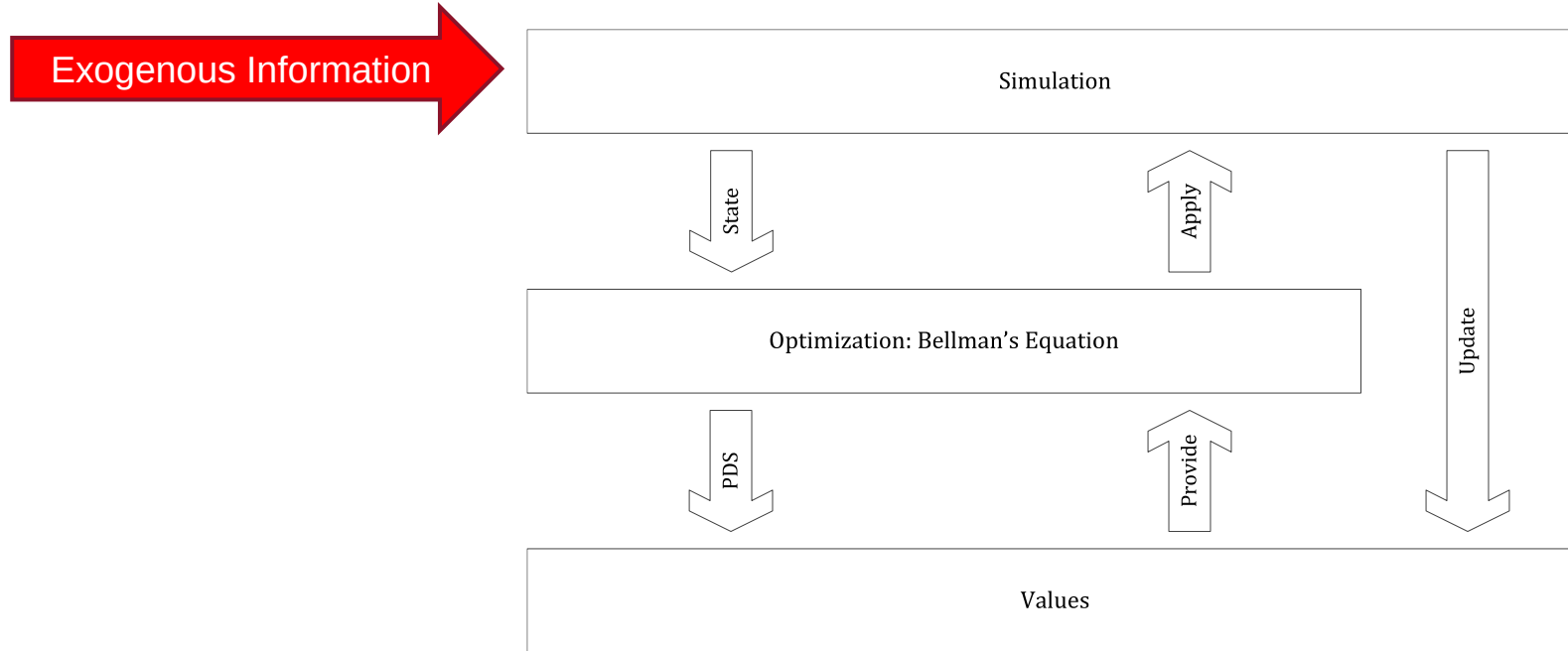
Attribute Scales

- Every attribute has exactly one level of measurement/scale
- Scale defines the types of operations and computations that are allowed for the attribute values

	Scale	Measure property	Mathematical operators	Example
Categorical	Nominal	Classification, membership	$=, \neq$	Color: red, green, blue, orange
	Ordinal	Comparison, level	$>, <$	Likert Scale: Like, Like Somewhat, Neutral, Dislike Somewhat, Dislike
Numerical / Cardinal	Interval	Difference, affinity	$+, -$	Date & Time: 05:00, 08:00, 11:00
	Ratio	Magnitude, amount	$\times, \div$	Salary: 25.000€, 50.000€, 150.000€

Berthold et al. (2010), Ch. 4.1

Exogenous Information



Exogeneous Information

- Little information suggests the use of policy function approx. or cost function approx.
- Anticipation may be achieved by explorative data analysis → derive future costs
- The more you know about exogeneous information, the more useful anticipation becomes
- Detailed exogeneous information suggests look-ahead methods or value function approx.
- Stochastic transitions denoted in the MDP are to be implemented in the simulation
- Transitions may be derived from distributions or they may be sampled from observations
- In the following, there are examples provided for both approaches

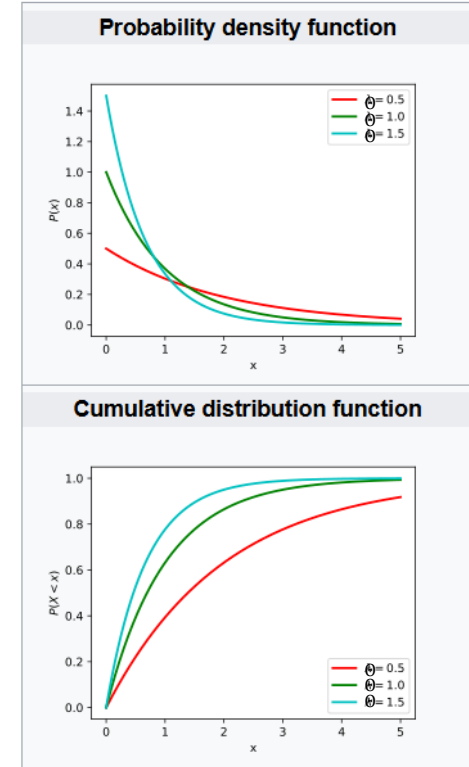
Generating stochastic transition from distribution

- Consider a stochastic transition depicting the elapse of time before the next customer arrives
- Inter-arrival times (between customers) are often depicted by an exponential distribution

$$X \sim \text{Exp}(\theta) : f(x) = \theta e^{-\theta x}$$

- Consider random variable with θ being the scale parameter
- Cumulative distribution function

$$\begin{aligned} P(X \leq x) &= F(x) = \int_{-\infty}^x f(t) dt \\ &= \int_0^x \theta e^{-\theta t} dt \\ &= -e^{-\theta t} \Big|_0^x \\ &= -(e^{-\theta x} - e^0) \\ &= 1 - e^{-\theta x} \end{aligned}$$



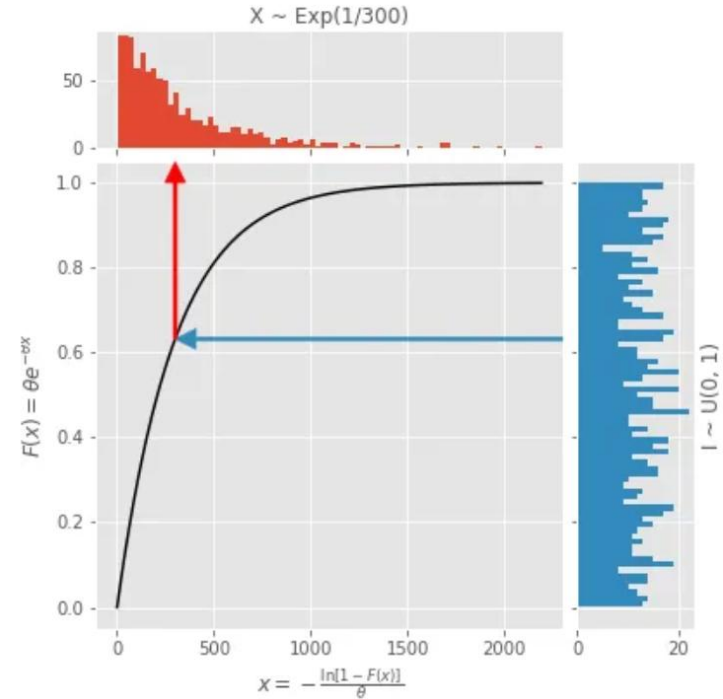
https://en.wikipedia.org/wiki/Exponential_distribution

Generating stochastic transition for distribution

- Apply inversion method if random variable comes with an invertible cumulative form F^{-1}
- Inversion of $F(x)$ for exponential distribution

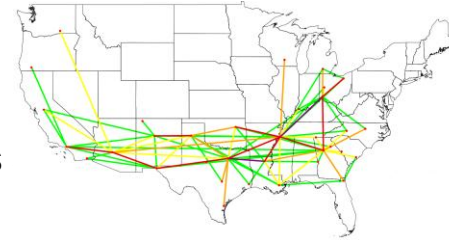
$$\begin{aligned}y &= 1 - e^{-\theta x} \\e^{-\theta x} &= 1 - y \\-\theta x &= \ln(1 - y) \\x &= -\frac{\ln(1 - y)}{\theta}\end{aligned}$$

- Generate random variable $X = F^{-1}(I \sim U(0, 1))$
- A sample from uniform distribution $U(0,1)$ is transposed into an exponentially dist. sample



Generating stochastic transition from observations

- A logistics problem is considered, which is to be solved with the help of sequential decision making. The problem extends over many decision points at which new commodities appear
- Commodities are to be delivered to their destination on time in a complex network.
- For this purpose, a simulation introduces commodities over time. A commodity has start location, a destination, a delivery time and a weight.
- All these features cannot simply be drawn from distributions.
- Since there is access to historic logistics data, new data can be generated for our problem by drawing samples from the historic pool of commodities at random

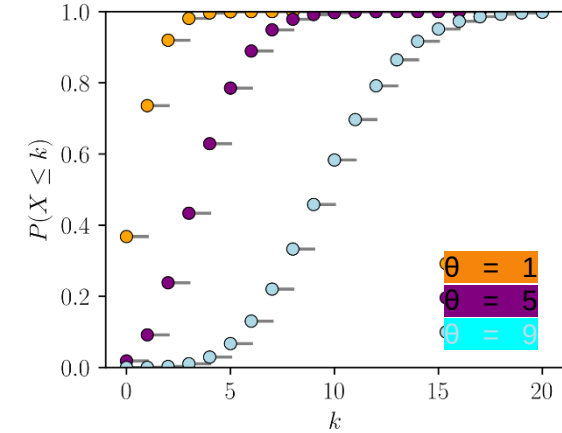
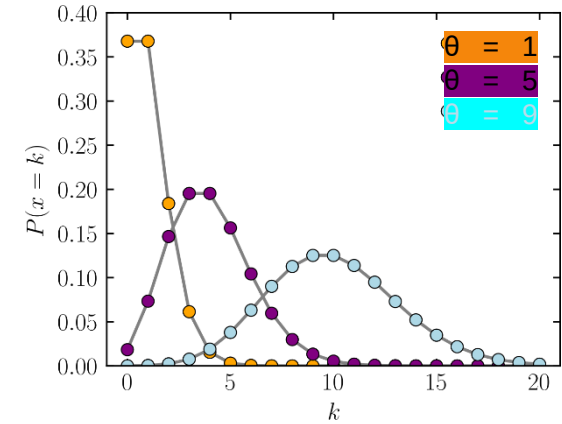


Choosing the number of commodities

- The first step is to decide how many new commodities will appear within each stochastic transition.
- In the literature, this number is usually determined using a discrete Poisson distribution (suggesting # of independent events occurring)

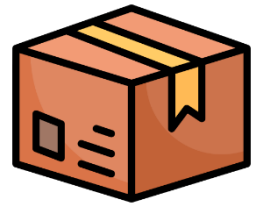
$$P_{\theta}(k) = \frac{\theta^k}{k!} e^{-\theta}$$

- Parameter θ determines the expected #, k denotes realizations.
- The CDF is built by summing up n probabilities $F_{\theta}(n) = \sum_{k=0}^n P_{\theta}(k)$
- Since the CDF is not invertible, draw x : $F_{\theta}(x) \leq U(0,1) \leq F_{\theta}(x+1)$



Choosing the commodities

- The second step is to select the commodities that appear in the respective transitions.
- As many commodities (as previously specified by the Poisson distribution) are drawn from the pool of all available commodities at each decision time.
- Since the (unknown) underlying distribution are already in the data, the selection can be made using a discrete uniform distribution.
- With this, each commodity x has the same probability of being selected from an array of n commodities ranging from 0 to $n - 1$: $x = \lfloor U(0,1) * n \rfloor$
- **In this way, a data set can be compiled for a sequential decision process.**

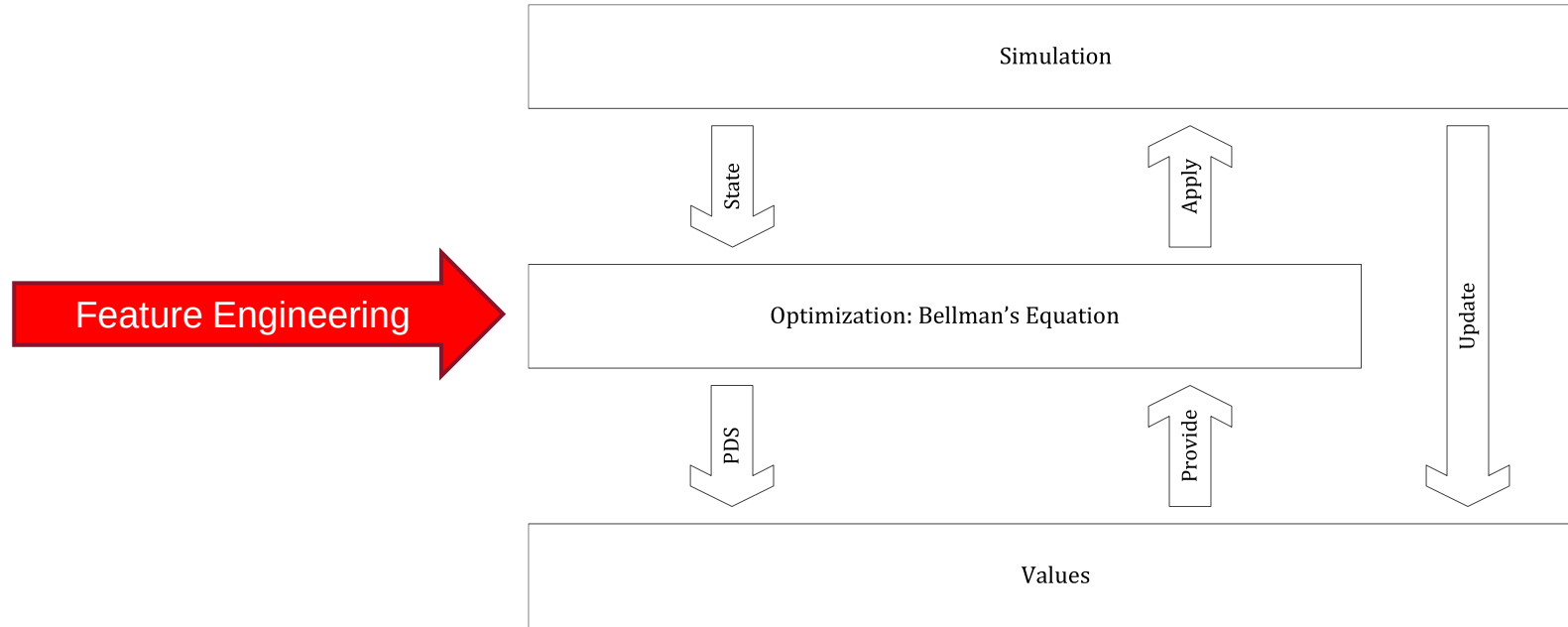


https://www.flaticon.com/de/kostenloses-iconpaket_7274757

Two issues with generating stochastic transitions

- Correlation
 - Random variables are not independent but follow some correlation
 - Correlation (due to causation) can appear time and/or space relation
 - For instance, a congestion on a road will cause a congestion on a parallel road
 - Zhaoxia Guo, Stein W. Wallace, Michal Kaut (2019) Vehicle Routing with Space- and Time-Correlated Stochastic Travel Times: Evaluating the Objective Function. INFORMS Journal on Computing 31(4):654-670
- Outlier
 - A distribution approximated from observations include historic peculiarities
 - In terms of a predictive model (what will happen?), overfitting is a topic
 - A predictive model comes with two objectives (accuracy and reduction of variance)
 - See Shmueli, G., & Koppius, O.R. (2011) Predictive Analytics in Information Systems Research. MIS Quarterly, 35(3), 553–572

Feature Engineering



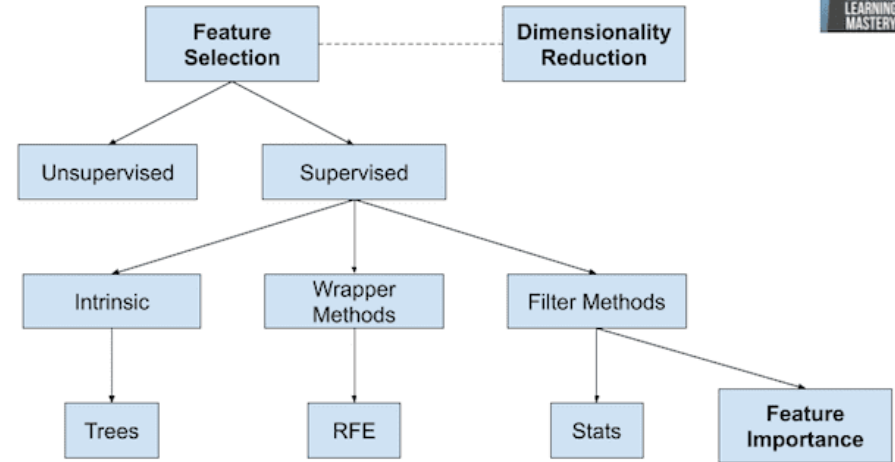
Decision Process

- Stochastic and dynamic decision modelling aims at generating an (near) optimal policy
- A policy provides a suitable decision to every state of the state space
- A small state- and decision space helps keeping anticipation computationally tractable
- Space reduction drives out dimensions from consideration (e.g. spatial or time)
- Next to problem specific dimension reduction generic analytics techniques can be applied
- Either important features are selected or the dimensionality of the state space is reduced

Feature selection techniques

- Unsupervised: do not incorporate variable
- Supervised: Determine features with strong relations to target variable by statistics
 - Intrinsic: For instance decision trees
 - Wrapper: Start with many features and successively eliminate features
 - Filter: Score features in accordance to relationship between feature and variable

Overview of Feature Selection Techniques



Copyright © MachineLearningMastery.com

<https://machinelearningmastery.com/feature-selection-with-real-and-categorical-data/>

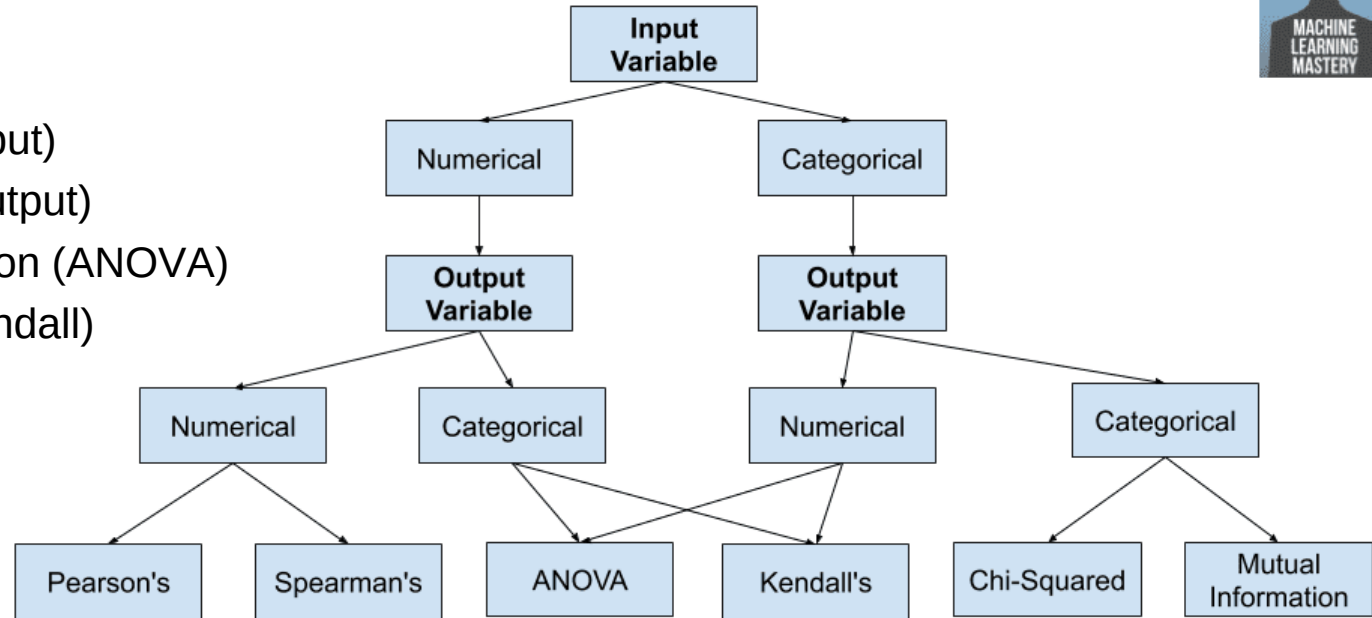


Filter based feature selection techniques

How to Choose a Feature Selection Method



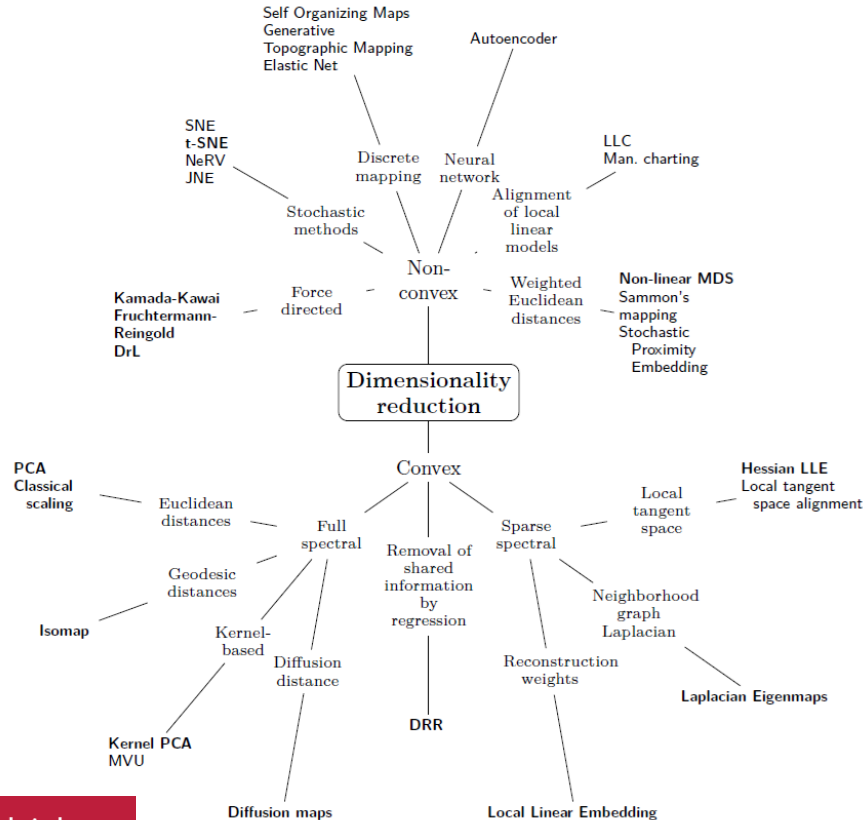
- Choice by
 - Feature type (input)
 - Variable type (output)
 - Normal distribution (ANOVA)
 - Rank based (Kendall)



<https://machinelearningmastery.com/feature-selection-with-real-and-categorical-data/>

Copyright © MachineLearningMastery.com

Taxonomy of methods for dimensionality reduction



Important properties



Parametrization

Parametric transformations allow the mapping of further data points into the lower-dimensional space



Linearity

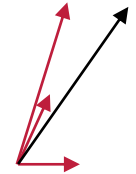
A linear transformation preserves linear properties of the data in the mapping



Convexity

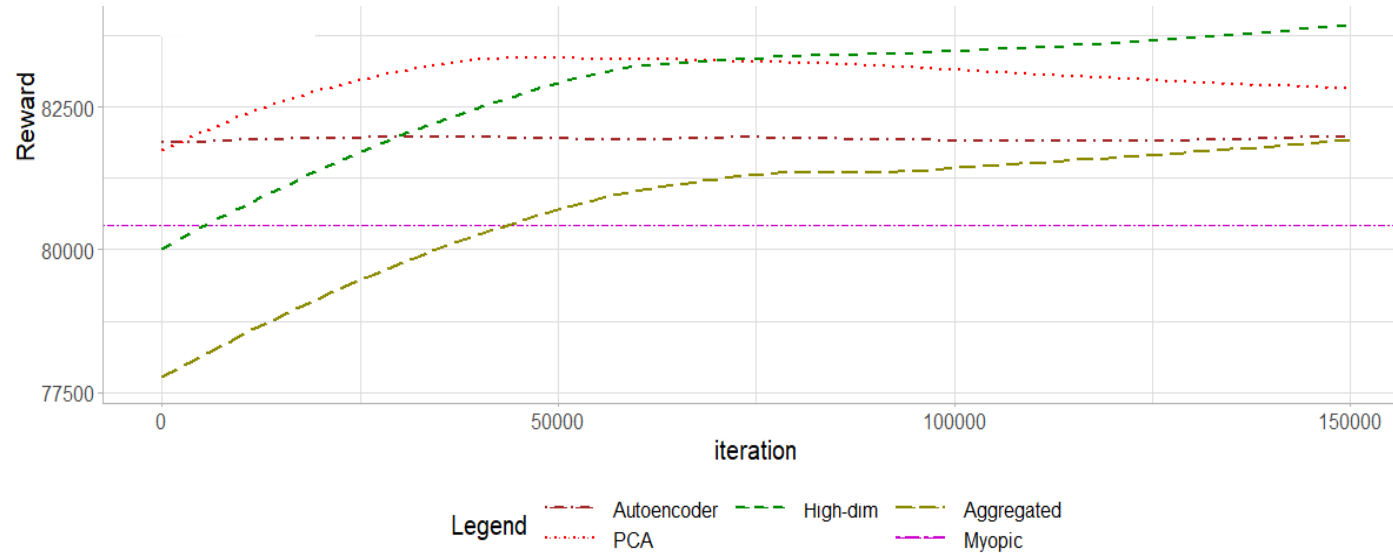
Convex objective functions guarantee the finding of the optimal low-dimensional space with respect to the information metric

Possible Candidates for Reduction of Dimensionality

Van der Maaten et al. (2009)	Method	Description	Graphik	
 Review paper testing dimensionality reduction methods on artificial and real datasets	PCA	- Lower-dimensional representation maximizing the variance of the data sample - Linear transformation - Parametric mapping		Accelerated learning With lowered dimensionality the combinatoric of states decreases which allows for more visits per state Reduced performance As dimensionality reduction is related to loss of information the performance in convergence is reduced
	Auto-encoder	- Lower-dimensional representation minimizing the minimal squared error of the reconstructed input - Non-linear transformation - Parametric mapping		

This and following slides: Dissertation of Julius Heitmann (2023)

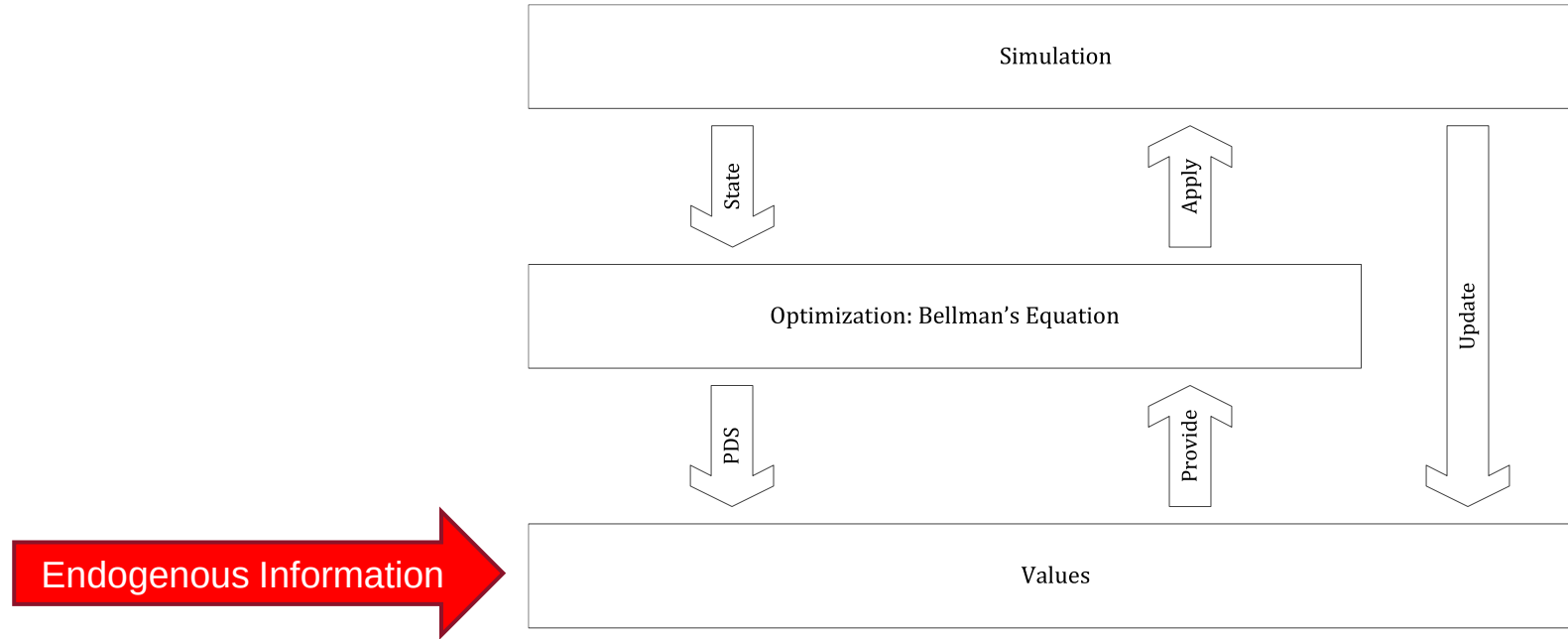
Performance on a demand responsive taxi problem



- PCA and AE learn (and perform) better at start
- In later iterations the high-dim model catches up

State representation	Relative performance	Accepted requests
High-dimensional state	104.3%	30.7
PCA state	103.1%	28.1
Autoencoder state	102.0%	31.3
Aggregated state	101.5%	29.8
Myopic reference	100.0%	30.6

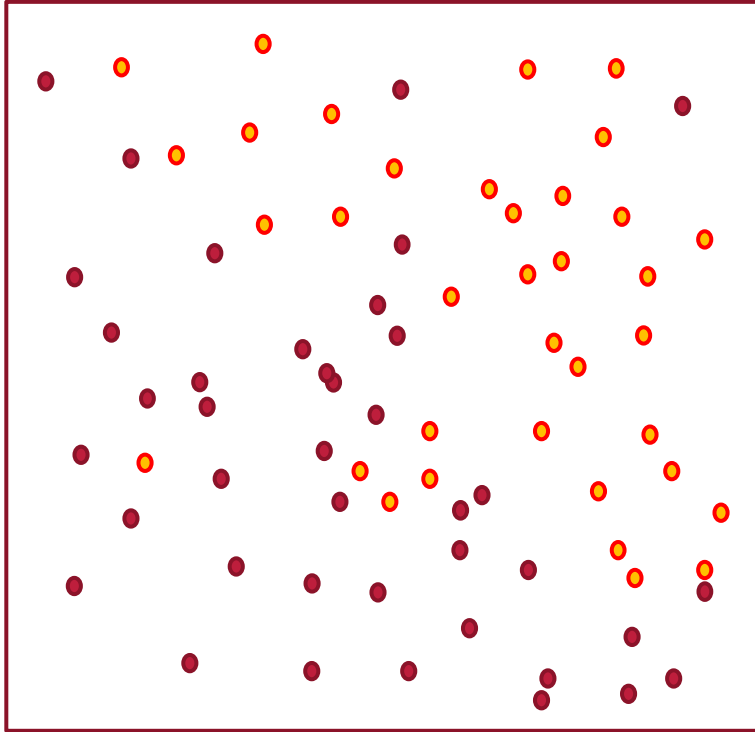
Endogenous information



Endogenous information

- Look-ahead approx. and value function approx. generate values from simulation runs
 - Values observed for multiple runs are aggregated into scalar values → data mining task
 - Supervised learning methods depicts relation between features and resulting value
 - Basically, we classify an value (output) upon observed features (inputs)
 - Classification (discrete) and regression (continuous) as well as ANN can be used.
-
- Supervised learning methods depend on feature and variable type (regression/classification)
 - White-box and black-box approaches can be distinguished (decision tree/ANN)
 - One design issue is the linearity of the underlying model (regression vs. ANN)
 - Methods differ in the number of observations required for learning (decision tree/ANN)

Classification: Example with 2-Dim Input and 2 Classes



Training samples

$$(x_1, y_1), \dots, (x_N, y_N) \in \mathbb{R}^2 \times \{0,1\}$$

Coordinate

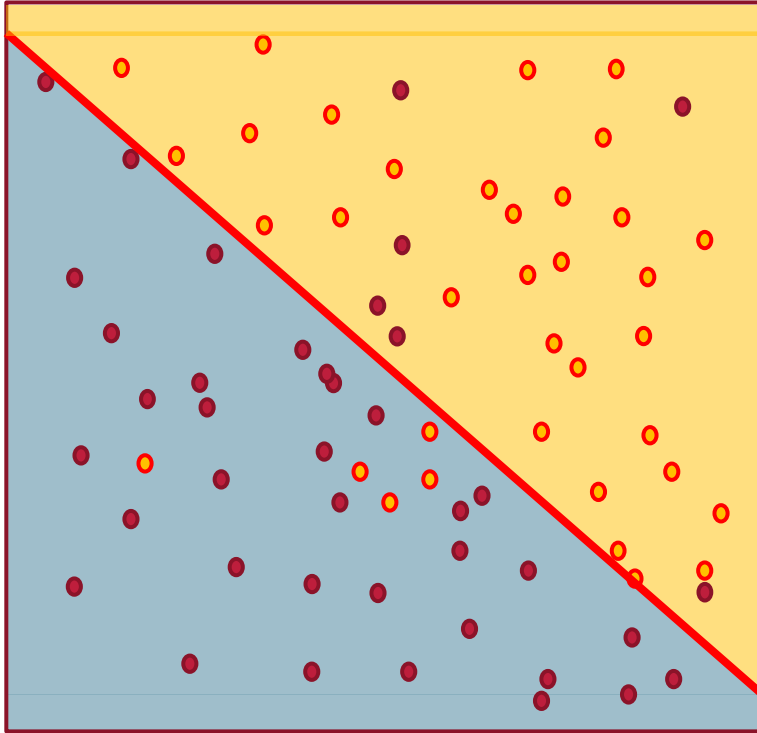
Class label

Task

Derive a hypothesis h that allows us to classify unlabeled data points

$$\hat{y} = h(x)$$

Classification: Linear Regression



Linear Regression

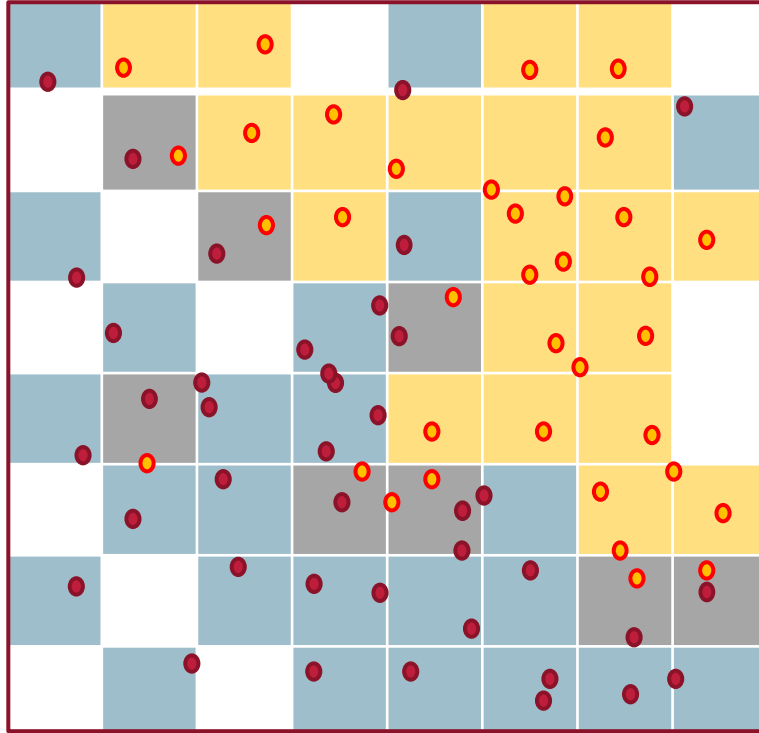
$$\hat{y} = \theta + w^T x$$

Score Bias Weight Input

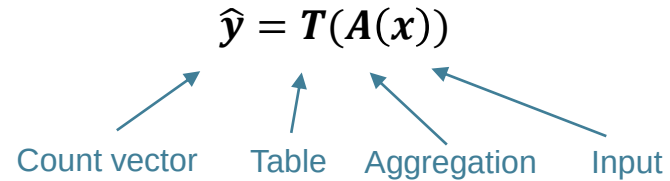
Using the score, we predict a class by

Orange if $\hat{y} > 0.5$
Red if $\hat{y} \leq 0.5$

Classification: Lookup-Table of Size 8x8



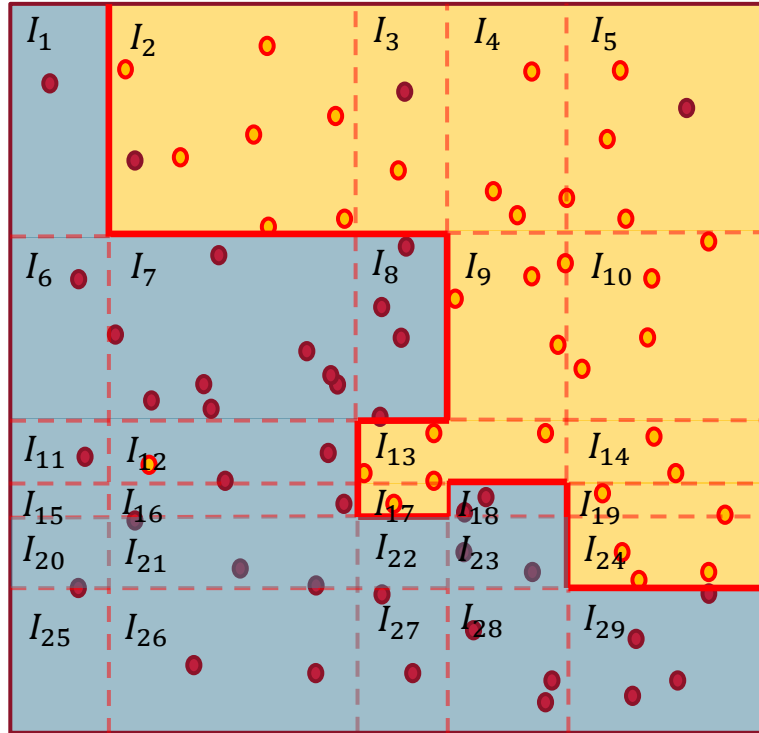
Table



A class is then predicted by

Orange if $\hat{y}_0 > \hat{y}_1$
Red if $\hat{y}_1 \leq \hat{y}_0$

Classification: Decision Tree with 29 Leaves



Decision Tree

$$\hat{y} = \sum_{k=1}^K p_k \cdot \mathbb{I}_{\{x \in I_k\}}$$

Probability
vector

Vector of class
probabilities

1 if x is in leaf I_k ,
0 else

A class is then predicted by

Orange if $\hat{y}_0 > \hat{y}_1$
Red if $\hat{y}_1 \leq \hat{y}_0$

Advanced Methods: Gradient Boosted Decision Trees

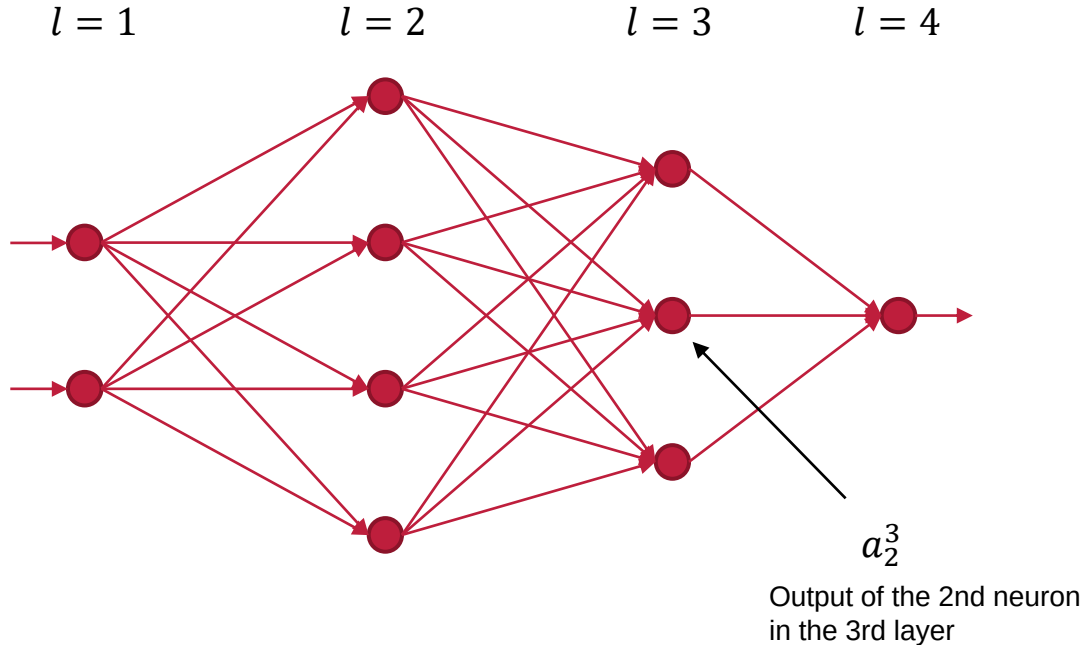
- **Reminder** A decision tree is noted by $F(x; p, I) := \sum_{k=1}^K p_k \cdot \mathbb{I}_{\{x \in I_k\}}$
- **Problem** A single decision tree is prone to overfitting
- **Solution** Fit several small trees and use them as an ensemble
- Iterative procedure $F_m(x) = F_{m-1}(x) + \underbrace{\alpha_m \sum_{j=1}^J p_{jm} \mathbb{I}_{\{x \in I_{jm}\}}}_{\text{Fit the residual of } F_{m-1}}$

Fit the residual of F_{m-1}

Advanced Methods: Neural Network

- **Reminder** Linear Regression $\hat{y} = \theta + w^T x$
- **Problem** Can only express linear relations
- **Solution** Stack several linear layers $\theta + w^T x$ and introduce non-linearity via an activation function, e.g., $ReLU(x) = \max(0, x)$

Advanced Methods: Neural Network



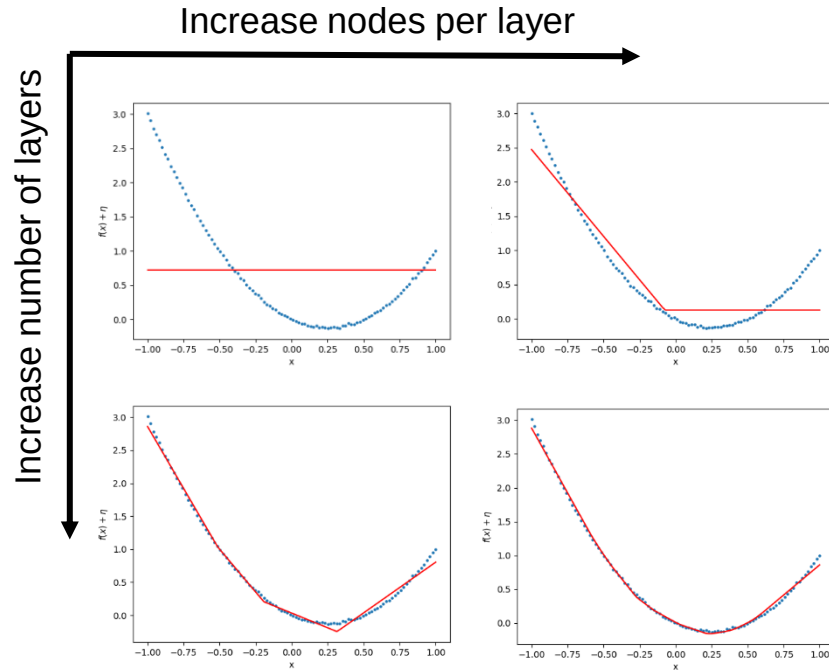
Output of a neuron

$$a_j^l = \sigma \left[\sum_k w_{jk}^l \cdot a_k^{l-1} + \theta_j^l \right]$$

Output of a layer

$$a^l = \sigma(w^l a^{l-1} + \theta^l)$$

Advanced Methods: Neural Network – Tuning



Also consider varying:

- Learning rate / Learning rate schedule
- Batch size
- Weight initialization
- Optimizer

Code available at:

https://github.com/flohilde/ifm_lecture

Some Words of Advice

- Start with a simple model to setup a benchmark. Then try out more complex models and compare them to your benchmark.
A possible progression: Linear regression → Gradient boosted decision tree → ANN
- When employing a complex method, train on a few thousand data points. It should be able to learn them by heart.
- Once you have a good model you can start fine-tuning. Only fine-tune hyperparameters that you actually understand and for which you can set reasonable ranges.
- Never resort to grid-search! Random-search and Bayesian optimization are much more sample efficient. Also, do use cross-validation.
- For neural networks: Architecture over hyperparameters!

Summary of lecture

- Simulation based methods (look-ahead, VFA) make use of descriptive analytics
- The size of the post-decision space of a problem can be lowered by dim. reduction
- Learn relation between features observed in the simulation and an expected future reward
- Generally, analytics provides generic approaches to design of ADP methods